

# Yongxiang Lei

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## Profile

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*Machine-learning researcher focusing on spatio-temporal forecasting and uncertainty quantification for renewable energy, process industry and climate applications (KAN, sequence models, diffusion, LSTM). Focusing on learning-based carbon-emission reduction tasks. Experienced in Unix-based data/compute pipelines, GPU-accelerated training, and operational deployment with engineering partners. Strong publication record and teaching/mentoring experience.*

## Research Interests

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- AI/DL/ML-driven environmental/climate/weather forecasting, interpretable neural operators (KAN), Mamba-style sequence models
- Probabilistic forecasting & uncertainty quantification, evaluation/benchmarking for decision support, LLM-based forecasting
- Faster, more efficient, and more stable AI forecasting pipelines/frameworks

## Employments

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**Research Assistant**, University of Warwick – Coventry, UK 01/2024 — 08/2025

- Studied on the integration of prediction theory with reinforcement learning.
- Worked on mathematical modeling and simulation and physiological feedback control mechanisms.
- Teaching Assistant in the lab session of an undergraduate feedback control class

**Graduate Teaching Assistant**, University of Warwick – Coventry, UK 2022 – Present

- ES4G4: Nonlinear Control and Reinforcement Learning
- ES197 : Systems Modelling, Simulation and Computation

**Software Engineer**, SmartMining – Changsha, China 07/2020 — 06/2021

- Core developer of Simulink Control Design Toolbox
- Led cross-team collaboration with Simulink Solvers and Physical Modeling

**Senior Controls Researcher**, ClearMotion – Milan, Italy 07/2019 — 01/2020

- Responsible for R&D of preview suspension control

**Principal Controls Engineer**, Pengcheng Laboratory – Shenzhen, China 01/2019 — 10/2019

- Led the controls team with 10 members across the worldwide.
- Improved the steering and speed control algorithms for passenger and commercial vehicles
- Integrated MBD (model-based design) into autonomous software development
- Developed robust control with tube model predictive control

## Education

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**University of Warwick**, 2nd PhD in Engineering 06/2024 – Present

- Major in reinforcement learning-based prediction and control
- Robust control
- Renewable and sustainable energy
- Wind turbine and wave energy converter
- Research topic was on Learning-Based Prediction and Control for Offshore Renewable and Sustainable Energy

**Politecnico di Milano (QS ranking 98)**, PhD in Mechanical Engineering (Control) 10/2020 – 06/2024

- Top 1 in the Mecc Department PhD Qualifying Exam
- Best PhD Dissertation award in the Mecc Department

- Honor in Mechanical Engineering at Department of Mechanical Engineering
- Dissertation topic was on Learning-Based Soft Sensing and Prediction for Industrial Applications
- Dissertation has been published as a book with Elsevier in 2025

**Central South University**, MS in Control Engineering

09/2017 – 06/2020

- GPA 88/100, ranked top 5% of department
- Visited the Politecnico di Milano to work on process control and machine learning

**University of South China**, BS in Automation

09/2013 – 06/2017

- Selected for National Key Class, a program for students achieving top scores in the national college entrance exam
- GPA 85/100, ranked top 10% of class

## Publications

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### Book

- [1] Y. **Lei** and R. K. Hamid, *Learning-based Soft Sensing and Prediction for Industrial Applications: Theory, Methodology and Application*. Elsevier, 2025.

### Weather/Climate-Based Forecasting

- [1] Y. **Lei**, B. Deng, and Z. Wang, “Climate temporal temperature prediction via an adaptive bidirectional residual mamba network,” Under Review.
- [2] Y. **Lei**, B. Deng, and Z. Wang, “DualMamba: Data-driven temporal climate forecasting via an enhanced bidirectional mamba network,” Under Review.

### AI/ML Relevant Forecasting

- [1] Z. Wang, J. Zheng, Y. **Lei**, T. Tao, K. Zuo, and W. Zhou, “Vmambamorph: A 3d multi-modality deformable image registration framework based on visual state space model with cross-scan module,” *ICASSP*, 2026.
- [2] C. Yang, X. Xiaojun, and Y. **Lei**, “Robust tube-based model predictive control for docking process self-balancing control of a reconfigurable unmanned ground vehicle,” *IEEE Transactions on Automation and Science Engineering*, 2026.
- [3] B. Deng, Y. **Lei**, and M. Li, “High-efficiency stochastic optimization for unit commitment with personalized demand response and wind power uncertainty,” *Computers & Industrial Engineering*, 2025.
- [4] Y. **Lei**, X. Chen, and Y. Xie, “Probabilistic temperature prediction in aluminum electrolysis using bayesian machine learning with uncertainty quantification,” *Engineering Applications of Artificial Intelligence*, 2025.
- [5] Y. **Lei**, X. Chen, Y. Yu, and A. E. Cetin, “Data-centric probabilistic temperature prediction with quantified uncertainty using bayesian machine learning for the aluminum electrolysis process,” *Control Engineering Practice*, 2025.
- [6] Y. **Lei**, B. Deng, and Z. Wang, “Climate temporal temperature prediction via an interpretable Kolmogorov-Arnold neural network,” *IEEE Transactions on Instrumentation and Measurement*, 2025.
- [7] Y. **Lei**, B. Deng, and Z. Wang, “Tem<sup>2</sup>-KAN: Data-driven temporal temperature prediction via an improved kolmogorov-arnold network,” *ISA Transactions*, 2025.
- [8] Y. **Lei** and H. R. Karimi, “Underflow concentration soft sensing for cone thickener system based on a direct data-driven quantile regression forecasting,” *Mechatronics Technology*, 2025.
- [9] Y. **Lei**, “A wave forecasting method based on probabilistic diffusion lstm network for model predictive control of wave energy converters,” *Applied Soft Computing*, 2024.
- [10] Y. **Lei**, “Intelligent optimal framework for the industrial mining plant-wide prediction control,” *IEEE Transactions on Instrumentation and Measurement*, 2023.
- [11] Y. **Lei** and H. R. Karimi, “DualLSTM: A novel key-quality prediction for a hierarchical cone thickener,” *Control Engineering Practice*, 2023.

- [12] Y. Lei and H. R. Karimi, "Underflow concentration prediction based on improved dual bidirectional lstm for hierarchical cone thickener system," *The International Journal of Advanced Manufacturing Technology*, 2023.
- [13] Y. Lei, X. Chen, Y. Xie, and L. Cen, "A self-supervised temporal temperature prediction method based on dilated contrastive learning," *Journal of Process Control*, 2022.
- [14] Y. Lei and H. R. Karimi, "A digital twin model of three-dimensional shading for simulation of the ironmaking process," *Machines*, 2022.
- [15] Y. Lei, H. R. Karimi, and X. Chen, "A novel self-supervised deep lstm network for industrial temperature prediction in aluminum processes application," *Neurocomputing*, 2022.
- [16] Y. Lei, F. Liu, H. R. Karimi, and X. Chen, "Manifold semi-supervised learning for aluminum electrolysis temperature identification based on regularized hierarchical extreme learning machine," *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, 2022.
- [17] N. Long, Y. Lei, L. Peng, P. Xu, P. Mao, *et al.*, "A scoping review on monitoring mental health using smart wearable devices," *Math. Biosci. Eng.*, 2022.
- [18] Y. Lei, X. Chen, and Y. Xie, "An improved cell situation identification approach with convolutional neural network and wavelet extreme learning machine," *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, 2021.
- [19] Y. Lei, H. R. Karimi, L. Cen, X. Chen, and Y. Xie, "Processes soft modeling based on stacked autoencoders and wavelet extreme learning machine for aluminum plant-wide application," *Control Engineering Practice*, 2021.
- [20] Y. Lei, X. Chen, M. Min, and Y. Xie, "A semi-supervised laplacian extreme learning machine and feature fusion with cnn for industrial superheat identification," *Neurocomputing*, 2020.
- [21] M. Min, X. Chen, Y. Lei, Z. Chen, and Y. Xie, "A novel kernel-based extreme learning machine with incremental hidden layer nodes," *IFAC-PapersOnLine*, 2020.
- [22] Y. Lei, L. Cen, X. Chen, and Y. Xie, "A hybrid regularization semi-supervised extreme learning machine method and its application," *IEEE Access*, 2019.

## Supervision

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| <b>Working experience with PI</b>                                                                                                                                                                                                | 2024 – Present |
| <ul style="list-style-type: none"> <li>• Dr. Hongyang Dong, Assistant Professor: University of Warwick</li> <li>• Dr. Jincheng Zhang, Assistant Professor (front): University of Warwick</li> </ul>                              |                |
| <b>Doctral Students</b>                                                                                                                                                                                                          | 2022 – Present |
| <ul style="list-style-type: none"> <li>• Congnan Yang: National University of Defense Technology</li> <li>• Yuechuan Xin : Shandong University of Science and Technology</li> <li>• Hanxuan Sun</li> </ul>                       |                |
| <b>Alumni with Theses or Publications</b>                                                                                                                                                                                        | 2023 – 2024    |
| <ul style="list-style-type: none"> <li>• Muhammad Shoaib, Master Thesis: Politecnico di Milano</li> <li>• Alice, Master Thesis: Politecnico di Milano</li> <li>• Mengcan Min, Master Thesis, Central South University</li> </ul> |                |

## Awards

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|------------------------------------------------------------------------------------------------------------------------|------|
| <b>Winner of the Oral Presentation</b>                                                                                 | 2025 |
| <ul style="list-style-type: none"> <li>• 12th Annual Postgraduate Research Showcase, University of Warwick.</li> </ul> |      |
| <b>Honor PhD Thesis</b>                                                                                                | 2024 |
| <ul style="list-style-type: none"> <li>• Politecnico di Milano, Department of Mechanics.</li> </ul>                    |      |
| <b>Excellent Reviewer</b>                                                                                              | 2020 |
| <ul style="list-style-type: none"> <li>• Journal of Guidance, Control, and Dynamics (JGCD).</li> </ul>                 |      |
| <b>Caitianxuanshu Young Author Prize</b>                                                                               | 2019 |
| <ul style="list-style-type: none"> <li>• Central South University, Changsha, China.</li> </ul>                         |      |

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| <b>Best MSc. Graduate Award</b>                                                                                                                                        | 2020 |
| <ul style="list-style-type: none"> <li>Outstanding (Top 1) performance on the control engineering Msc. Qualifying Examination, School of Automation at CSU.</li> </ul> |      |
| <b>National First Place</b>                                                                                                                                            | 2021 |
| <ul style="list-style-type: none"> <li>The Chinese Graduate Mathematical Contest in Modeling.</li> </ul>                                                               |      |
| <b>National First Place</b>                                                                                                                                            | 2020 |
| <ul style="list-style-type: none"> <li>The Chinese Undergraduate Mathematical Contest in Modeling</li> </ul>                                                           |      |

## Services

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### Guested Editor for Journals

*Special Issue "Time Series Forecasting and Soft Sensing for Climate Modeling and Applications", Applied Sciences*

- Machine learning and deep learning approaches for climate time series prediction
- Uncertainty quantification and interpretability in climate forecasting models
- Case studies on real-world climate data and large-scale simulations
- Edge and cloud computing for climate data analysis and sensing
- Soft sensing methods for estimating unmeasured or hard-to-measure climate variables
- Hybrid models combining physical and data-driven methods

*Special Issue "Advanced Prediction and Control for Offshore Energy: Harvesting, Conversion, Storage and Environmental Sustainability", Energy Engineering*

- Resource characterization and short-/long-term forecasting of offshore wind, wave, tidal and current energy using numerical models, remote sensing and data-driven methods
- Advanced control strategies for fixed and floating offshore wind turbines, wave and tidal energy converters, and hybrid offshore energy systems
- Model predictive control, robust and adaptive control, and nonlinear control for load mitigation, power maximization and fatigue-life extension in harsh marine environments
- Data-driven and AI-based approaches (machine learning, deep learning, reinforcement learning) for forecasting, control, fault diagnosis, and decision support in offshore energy systems
- Integrated prediction–control frameworks for co-optimizing structural design, mooring systems, platform dynamics, and power conversion

### Reviewer for Conferences and Journals

- Conferences: American Control Conference; Conferences on Decision and Control
- Journals: International Journal of Forecasting; Applied Soft Computing; IEEE Transactions on Automatic Control (TAC); Automatica; IEEE Transactions on Neural Network and Learning Systems (TNNLS); Springer Nature

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| <b>Early Career Advisory Board</b>                                                                    | 2022 – Present |
| <ul style="list-style-type: none"> <li>Control Engineering Practice (CEP), an IFAC Journal</li> </ul> |                |

## Technical Skills

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**Programming:** Python, C++ , JavaScript, Fortran, MATLAB

**Frameworks/Libraries:** PyTorch, TensorFlow, NumPy, SciPy, Pandas, Flask, Django, React, OpenFAST

**DevOps & Tools:** VS code, Git/GitHub, GitHub Actions (CI/CD), Docker, Kubernetes, HPC clusters

**Climate Databases:** UK Met Office, MySQL, MongoDB

**Documentation & Testing:** Sphinx, ReadTheDocs, pytest